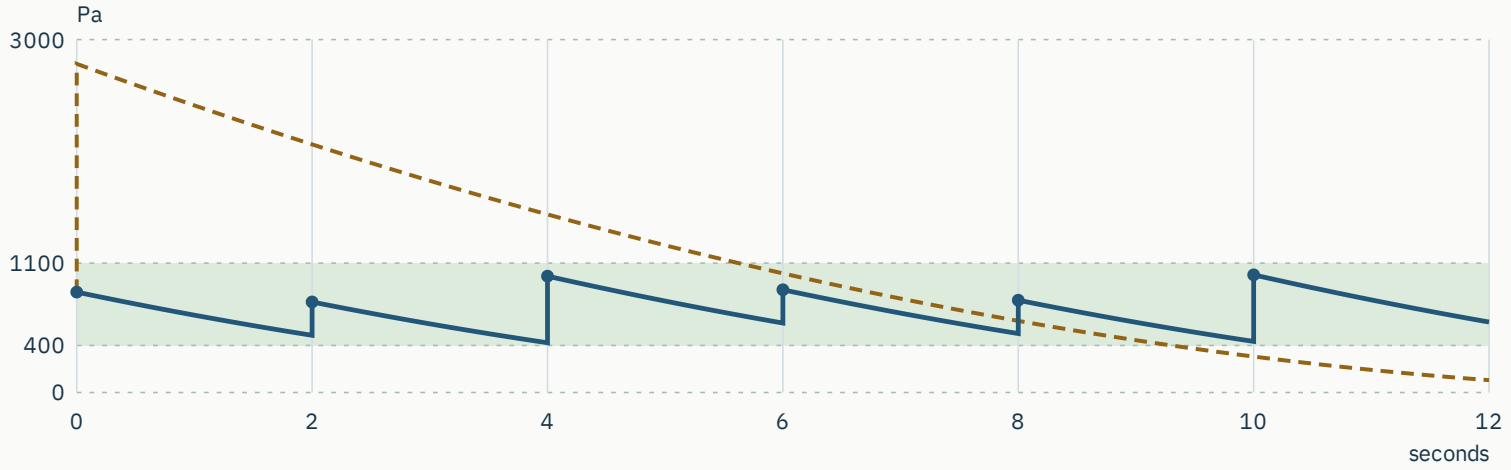
Seven strokes. Twelve seconds.

Modern timing exercise for Pnigeus. The target of 400–1100 Pa is chosen for this worksheet, not measured from ancient pipes. Three starting strokes precede the clock. Each exercise stroke delivers one ambient-equivalent litre instantaneously at the start of its slot. Pipes remain open throughout.



Solid blue: current. Dashed brown, if present: kept comparison. Green band: the chosen target. Both traces use the same pressure and time scales.

Current timing

2 open pipes; tank area 6 × chamber area; strokes at 0, 2, 4, 6, 8, 10 s: 0, 1, 2, 1, 1, 2. 7 of seven exercise strokes placed, in addition to three starting strokes.

Below target: 0.000000 s. Within: 12.000000 s. Above: 0.000000 s. Air through pipes: 7.899693 ambient L. Vented: 0.000000 ambient L.

Current timing: 2 open pipes; tank area 6 × chamber area. Pressures bracket each two-second release.

Time, s	Strokes	Start, Pa	End, Pa	Below, s	Within, s	Above, s
0–2	0	853.597	486.216	0.000000	2.000000	0.000000
2–4	1	769.660	422.986	0.000000	2.000000	0.000000
4–6	2	989.034	590.447	0.000000	2.000000	0.000000
6–8	1	873.332	501.244	0.000000	2.000000	0.000000
8–10	1	784.608	434.160	0.000000	2.000000	0.000000
10–12	2	1000.089	599.061	0.000000	2.000000	0.000000

Kept comparison

2 open pipes; tank area 6 × chamber area; strokes at 0, 2, 4, 6, 8, 10 s: 7, 0, 0, 0, 0, 0. 7 of seven exercise strokes placed, in addition to three starting strokes.

Below target: 2.698894 s. Within: 3.678777 s. Above: 5.622329 s. Air through pipes: 9.633315 ambient L. Vented: 0.000000 ambient L.

Kept timing: 2 open pipes; tank area 6 × chamber area. Pressures bracket each two-second release.

Time, s	Strokes	Start, Pa	End, Pa	Below, s	Within, s	Above, s
0–2	7	2793.595	2108.933	0.000000	0.000000	2.000000
2–4	0	2108.933	1513.930	0.000000	0.000000	2.000000
4–6	0	1513.930	1012.711	0.000000	0.377671	1.622329
6–8	0	1012.711	608.914	0.000000	2.000000	0.000000
8–10	0	608.914	305.586	0.698894	1.301106	0.000000
10–12	0	305.586	105.085	2.000000	0.000000	0.000000

My explanation

```
Both timings receive ten ambient-equivalent litres: three before the clock, then
seven placed strokes. Giving all seven at the start raises the model outlet rate and
consumes more gas early. With the same pipe and tank settings, moving deliveries
later keeps the pressure inside the chosen band for all twelve seconds. This is a
pressure account under the stated teaching assumptions, not a test of actual organ
pipes.
```

Reopen the current timing. The kept comparison is retained in this file; the link opens only the current timing.

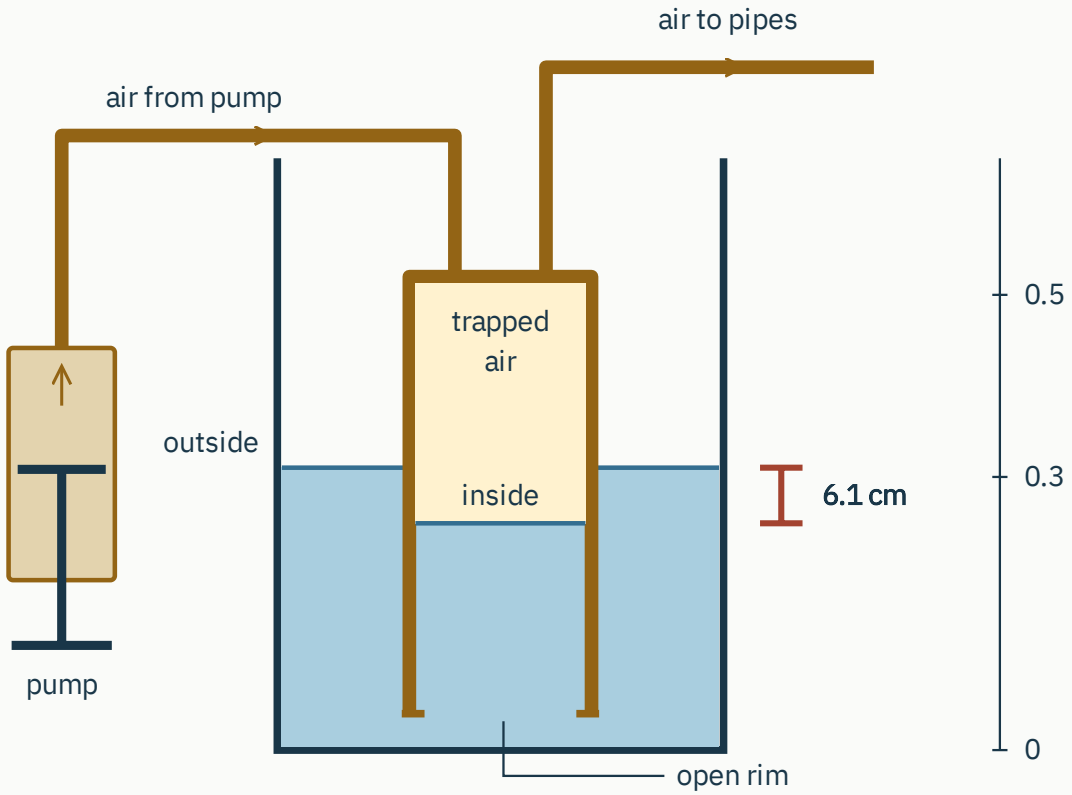
Download interval account CSV · Download full action account CSV

Time within the band is integrated at threshold crossings, not counted from the graph's samples. The window measures pressure, not audible sound quality. There is no claim that this would sustain real pipes. Finite pumping time, effort, fluid inertia and acoustics are omitted.

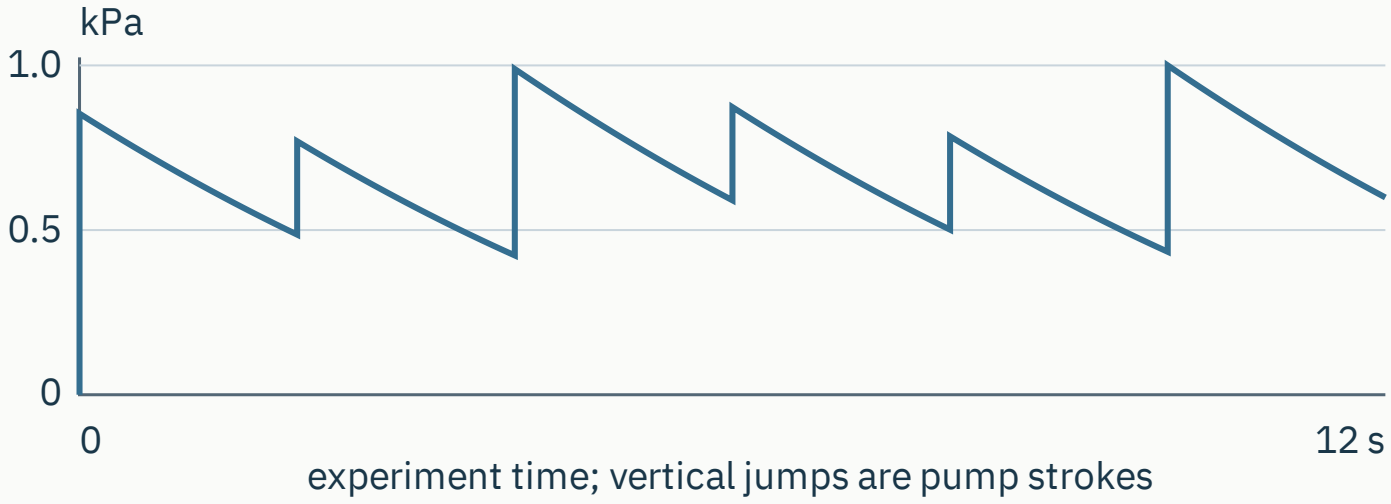
The water inside a water organ

Pnigeus · saved modern experiment · tank area 6 × chamber area · experiment time 12 s.

Gauge pressure 599.061 Pa; head 6.109 cm. Supplied 10.000 ambient-equivalent litres, released 7.900, vented 0.000. The chamber still holds 10.836 actual litres of air.



Modern constant-section model; vertical scale in metres



Solid blue: water chamber. The dry comparison is hidden in this graph; its numerical results remain in the action table.

Reopen this exact action sequence. The reading, image, fonts and audio below are embedded and need no connection. Linked external editions require a connection.

Last release

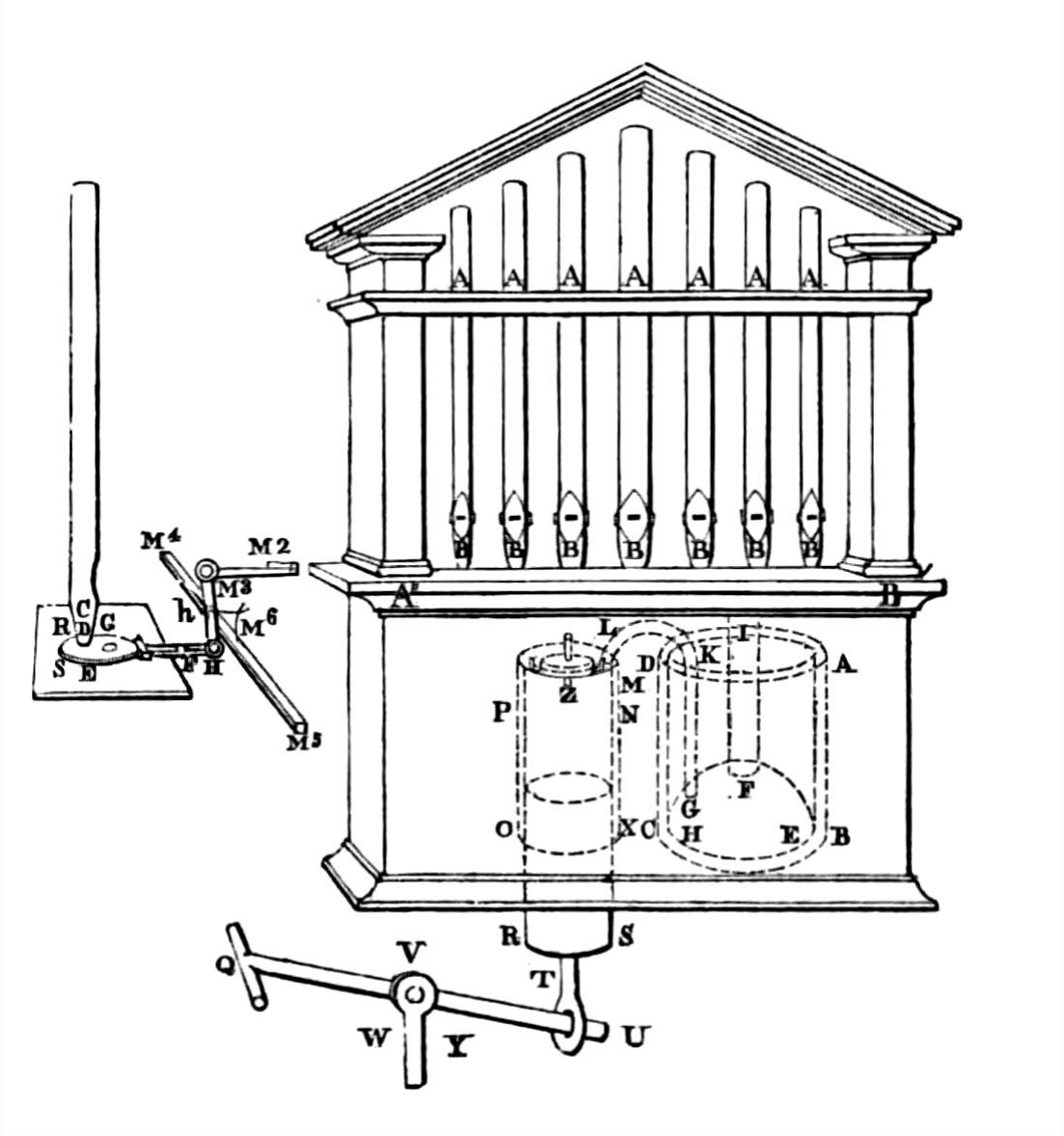
Action 17: Release for 2 s. Modern sine reference tones; illustrative pressure-to-amplitude mapping.

Actions in order

Action	Command	Time, s	Water, Pa	Dry, Pa	Vented, L
1	Pump 1 litre	0	286.082	11514.205	0.000000
2	Pump 1 litre	0	570.606	23028.409	0.000000
3	Pump 1 litre	0	853.597	34542.614	0.000000
4	Open 2 pipes (gates 3)	0	853.597	34542.614	0.000000
5	Release for 2 s	2	486.216	0.000	0.000000
6	Pump 1 litre	2	769.660	11514.205	0.000000
7	Release for 2 s	4	422.986	0.000	0.000000
8	Pump 1 litre	4	706.771	11514.205	0.000000
9	Pump 1 litre	4	989.034	23028.409	0.000000
10	Release for 2 s	6	590.447	0.000	0.000000
11	Pump 1 litre	6	873.332	11514.205	0.000000
12	Release for 2 s	8	501.244	0.000	0.000000
13	Pump 1 litre	8	784.608	11514.205	0.000000
14	Release for 2 s	10	434.160	0.000	0.000000
15	Pump 1 litre	10	717.885	11514.205	0.000000
16	Pump 1 litre	10	1000.089	23028.409	0.000000
17	Release for 2 s	12	599.061	0.000	0.000000

Read the device

Vitruvius writes in Latin; the chamber’s name is borrowed from Greek. His description makes a useful reading exercise: find the container, the water and the object turned upside down.



Hero, *Pneumatics*, section 76, fig. 76. Greenwood's translation, edited by Woodcroft (1851), pp. 105–107. This is a nineteenth-century explanatory drawing, not a surviving ancient plan. Read the full description.

Vitruvius, *De architectura* X.8.2

intra arcam, quo loci aqua sustinetur, inest pnigeus uti infundibulum inversum

Inside the chest, where the water is held, there is a pnigeus, like an inverted funnel.

New English translation of this excerpt. The Latin word notes below explain the grammar and identify the corresponding parts.

Inspect Rose's Latin edition (1899). This pump model uses a straight-sided chamber. Hero calls his a hollow hemisphere; Vitruvius's funnel comparison need not mean a cone. Read Hero's Greek and explore chamber shape, including Hagel's discussion of rounded Roman funnels.

Greek words at the bench

πνιγεύς pnigeus

The submerged chamber's name in this description. Other contexts give this word other senses; this is not a universal word for "air tank."

ὕδωρ hydōr
Water. Its combining form *hydr-* survives in "hydraulic." Here it helps hold the supply of air.

ἀήρ aēr
Air. This is what reaches the pipes. The English word "air" shares its ancestry.

Greek headwords and transliterations are reading aids, not a reconstructed pronunciation. πνιγεύς in LSJ; ὕδωρ; ἀήρ.

Latin word notes

intra arcam: 'inside the chest'. Intra takes the accusative; arcam is the accusative singular of arca. The bronze chest contains the water.

aqua sustinetur: 'water is held'. Aqua is the nominative subject; sustinetur is a present passive verb. The drawing's blue regions are connected under the rim.

pnigeus: the chamber's Greek loanword, nominative subject of inest ('is inside'). Its air is connected to the pump and the pipes, while its open lower edge lets water move.

infundibulum inversum: 'an inverted funnel'. Inversum agrees with the neuter infundibulum. The shape is Vitruvius's comparison, not the constant cross-section chosen for this experiment.

What the experiment assumes

The ancient descriptions supply a pump, an inverted chamber, water and selectively opened pipes. Everything numerical here belongs to a modern, quasi-static teaching model. Water settles instantly; air has constant temperature. We omit wave motion, pump mechanics, heat, leakage, wind-chest losses and actual pipe acoustics.

Two surfaces, one water supply

The chamber has a constant area of 0.04 m², its roof is 0.52 m high, its rim 0.04 m high. At rest both surfaces are 0.30 m high. If the inside surface drops by d metres, the outside rises by d/(R − 1), where R is the total tank area divided by the chamber area.

```
head = d × R / (R − 1)
gauge pressure = ρ × g × head
air volume = 8.8 + 40d litres
```

We use ρ = 1000 kg/m³, g = 9.80665 m/s² and atmospheric pressure 101325 Pa. The head is the difference between the two surfaces, not their height above the floor. Hydrostatic pressure, NASA.

Count the air, then let it out

Air stock is measured as the volume that the same amount of gas would occupy at our atmospheric pressure and temperature. One stroke delivers one of these ambient-equivalent litres. The chamber's actual air volume is different when pressurized.

```
stock = air volume × (1 + pressure / 101325)
flow = open pipes × 0.4 × √(pressure / 1000)
                          ambient litres / second
```

The flow law and coefficient are chosen. An exact time integral gives the next settled state; it stops withdrawing air at zero gauge pressure. Once inside water reaches the open rim, further delivered air escapes. No water is consumed.

▼ Why compare a sealed dry chamber?

It starts with the same 8.8 litres of air at atmospheric pressure and receives the same strokes and gate commands. Its volume stays fixed, its air is isothermal and it has no relief outlet. It is a separate counterfactual vessel, not the ancient chamber with its water removed. Its pressure is 101325 × (stock/8.8 − 1). A one-litre delivery produces a much larger pressure rise and the same outlet rule then removes air faster. The shared graph uses one pressure scale; the water trace can therefore appear small.

▼ What does the sound establish?

Four modern sine test tones form an A-major chord: A3, C♯4, E4 and A4 at A4 = 440 Hz. Their frequencies stay fixed. Amplitude follows an illustrative square-root pressure mapping, with short click-suppressing edges. These sounds mark the model's air reserve; they do not reconstruct ancient tuning, pipe timbre, loudness, voicing or pressure-dependent pitch. Live playback and WAV downloads use the same finite samples.

The experiment clock advances only with "Release." A pump stroke is an idealized instantaneous delivery. Changing the tank replays all actions with its new geometry. Shared links and downloads record the action order, including gate changes made at the same experiment time.

Font notices


```
Copyright © 2017 IBM Corp. with Reserved Font Name "Plex"

This Font Software is licensed under the SIL Open Font License, Version 1.1.
```

This license is copied below, and is also available with a FAQ at: http://scripts.sil.org/OFL

SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007

PREAMBLE

The goals of the Open Font License (OFL) are to stimulate worldwide development of collaborative font projects, to support the font creation efforts of academic and linguistic communities, and to provide a free and open framework in which fonts may be shared and improved in partnership with others.

The OFL allows the licensed fonts to be used, studied, modified and redistributed freely as long as they are not sold by themselves. The fonts, including any derivative works, can be bundled, embedded, redistributed and/or sold with any software provided that any reserved names are not used by derivative works. The fonts and derivatives, however, cannot be released under any other type of license. The requirement for fonts to remain under this license does not apply to any document created using the fonts or their derivatives.

DEFINITIONS

"Font Software" refers to the set of files released by the Copyright Holder(s) under this license and clearly marked as such. This may include source files, build scripts and documentation.

"Reserved Font Name" refers to any names specified as such after the copyright statement(s).

"Original Version" refers to the collection of Font Software components as distributed by the Copyright Holder(s).

"Modified Version" refers to any derivative made by adding to, deleting, or substituting -- in part or in whole -- any of the components of the Original Version, by changing formats or by porting the Font Software to a new environment.

"Author" refers to any designer, engineer, programmer, technical writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS

Permission is hereby granted, free of charge, to any person obtaining a copy of the Font Software, to use, study, copy, merge, embed, modify, redistribute, and sell modified and unmodified copies of the Font Software, subject to the following conditions:

1) Neither the Font Software nor any of its individual components, in Original or Modified Versions, may be sold by itself.

2) Original or Modified Versions of the Font Software may be bundled, redistributed and/or sold with any software, provided that each copy contains the above copyright notice and this license. These can be included either as stand-alone text files, human-readable headers or in the appropriate machine-readable metadata fields within text or binary files as long as those fields can be easily viewed by the user.

3) No Modified Version of the Font Software may use the Reserved Font Name(s) unless explicit written permission is granted by the corresponding

Copyright Holder. This restriction only applies to the primary font name as
presented to the users.

4) The name(s) of the Copyright Holder(s) or the Author(s) of the Font
Software shall not be used to promote, endorse or advertise any
Modified Version, except to acknowledge the contribution(s) of the
Copyright Holder(s) and the Author(s) or with their explicit written
permission.

5) The Font Software, modified or unmodified, in part or in whole,
must be distributed entirely under this license, and must not be
distributed under any other license. The requirement for fonts to
remain under this license does not apply to any document created
using the Font Software.

TERMINATION
This license becomes null and void if any of the above conditions are
not met.

DISCLAIMER
THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND,
EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF
MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT
OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE
COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY,
INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL
DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM
OTHER DEALINGS IN THE FONT SOFTWARE.

Copyright 2022 The Noto Project Authors (https://github.com/notofonts/latin-greek-cyrillic)

This Font Software is licensed under the SIL Open Font License, Version 1.1.
This license is copied below, and is also available with a FAQ at:
https://scripts.sil.org/OFL

SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007

PREAMBLE
The goals of the Open Font License (OFL) are to stimulate worldwide
development of collaborative font projects, to support the font creation
efforts of academic and linguistic communities, and to provide a free and
open framework in which fonts may be shared and improved in partnership
with others.

The OFL allows the licensed fonts to be used, studied, modified and
redistributed freely as long as they are not sold by themselves. The
fonts, including any derivative works, can be bundled, embedded,
redistributed and/or sold with any software provided that any reserved
names are not used by derivative works. The fonts and derivatives,
however, cannot be released under any other type of license. The
requirement for fonts to remain under this license does not apply
to any document created using the fonts or their derivatives.

DEFINITIONS
"Font Software" refers to the set of files released by the Copyright

Holder(s) under this license and clearly marked as such. This may include source files, build scripts and documentation.

"Reserved Font Name" refers to any names specified as such after the copyright statement(s).

"Original Version" refers to the collection of Font Software components as distributed by the Copyright Holder(s).

"Modified Version" refers to any derivative made by adding to, deleting, or substituting -- in part or in whole -- any of the components of the Original Version, by changing formats or by porting the Font Software to a new environment.

"Author" refers to any designer, engineer, programmer, technical writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS
Permission is hereby granted, free of charge, to any person obtaining a copy of the Font Software, to use, study, copy, merge, embed, modify, redistribute, and sell modified and unmodified copies of the Font Software, subject to the following conditions:

1) Neither the Font Software nor any of its individual components, in Original or Modified Versions, may be sold by itself.

2) Original or Modified Versions of the Font Software may be bundled, redistributed and/or sold with any software, provided that each copy contains the above copyright notice and this license. These can be included either as stand-alone text files, human-readable headers or in the appropriate machine-readable metadata fields within text or binary files as long as those fields can be easily viewed by the user.

3) No Modified Version of the Font Software may use the Reserved Font Name(s) unless explicit written permission is granted by the corresponding Copyright Holder. This restriction only applies to the primary font name as presented to the users.

4) The name(s) of the Copyright Holder(s) or the Author(s) of the Font Software shall not be used to promote, endorse or advertise any Modified Version, except to acknowledge the contribution(s) of the Copyright Holder(s) and the Author(s) or with their explicit written permission.

5) The Font Software, modified or unmodified, in part or in whole, must be distributed entirely under this license, and must not be distributed under any other license. The requirement for fonts to remain under this license does not apply to any document created using the Font Software.

TERMINATION
This license becomes null and void if any of the above conditions are not met.

DISCLAIMER
THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY,

```
INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL
DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING
FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM
OTHER DEALINGS IN THE FONT SOFTWARE.
```

Original code, diagrams and English explanations: PolyForm Noncommercial, © 2026 Anthony Mercer. The 1851 source illustration is public domain in the United States. Edition af49cdf0dc103cdda07aec2926a8c33921465a4c63e132909bacdc68d41e5875.